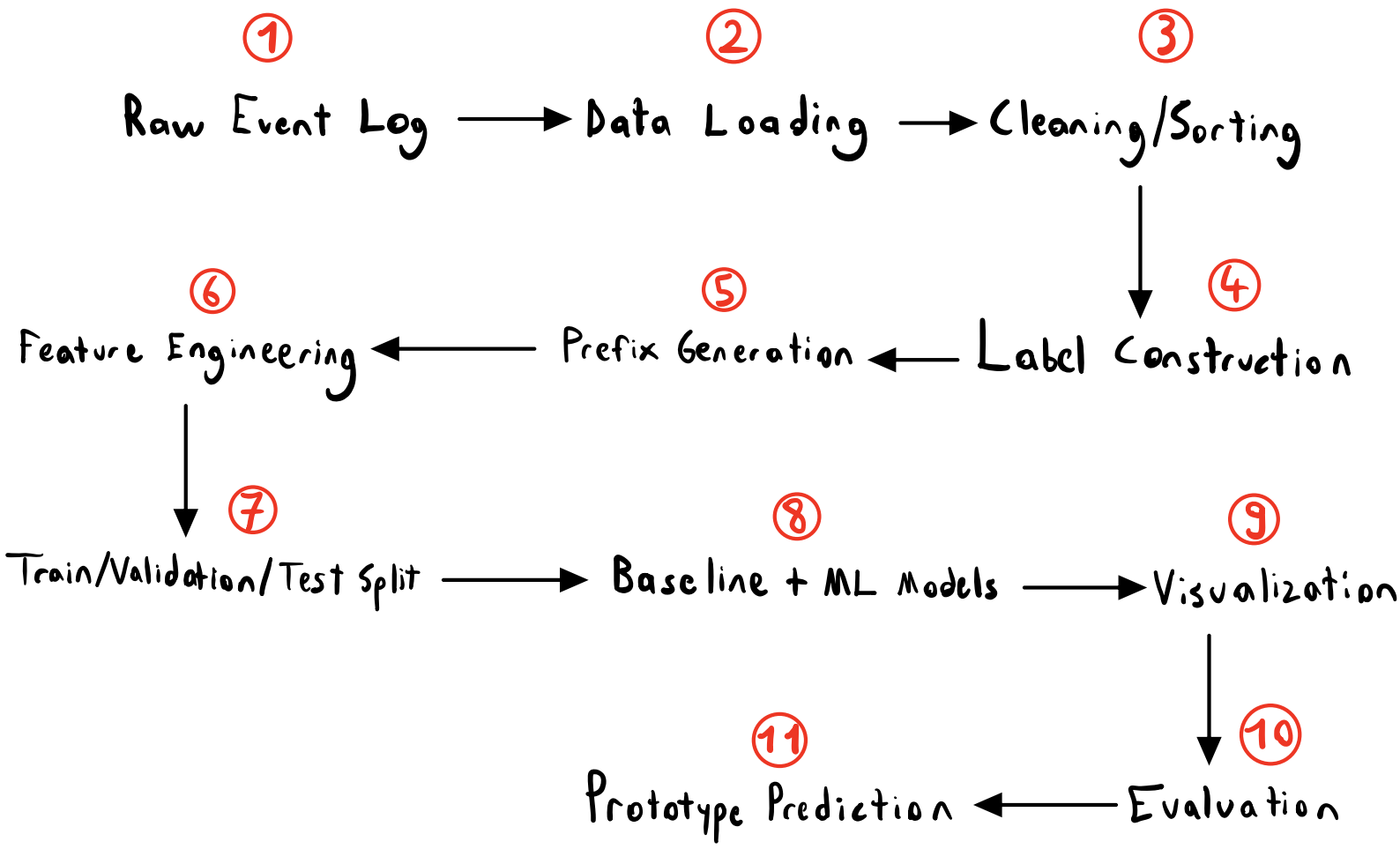

1 Store the original event log.

2 Load the data. If CSV → pandas
XES → PM4Py.
Standardize column names.
Convert the timestamp column to the datetime format.

3 Sort the events in each case by time.
Remove any duplicate events.
Check for negative or invalid time sequences.
Fix timestamp time zone issues.
} pandas

4. Compute case completion time (last event).
For each prefix: remaining_time = completion time - last event time
Use consistent unit (hours or days).
Create column: remaining time

pandas

5. Convert each case into multiple prefixes.
Each prefix → one training sample.
Transforms data from case-level → prefix-level.
Target label = remaining time

pandas

6. Create features from prefixes:

Aggreted dynamic features
- prefix length
- elapsed time
- activity counts / last activity

Static attributes (priority)
temporal features (day / hour)
Encode categorical features

pandas, Scikit-Learn

7. Split data by case and time (%70 train, %15 validation, %15 test)
Avoid data leakage!

pandas, Scikit-Learn

8. Baseline: mean remaining time
ML Models: Linear Regression, Random Forest, etc.
Tune hyperparameters using validation set.

Scikit-Learn

9 – 10
Evaluate on test set using:
- MAE
- RMSE

Analyze performance by prefix length.
Create plots.

pandas, Scikit-Learn, Matplotlib

11 Load trained model

Apply same preprocessing + features.

Predict remaining time for new running cases.

Provide script or notebook for demo.

pandas, Scikit-Learn, PM4Py